

7. Answer: C

Let the resistive force be D and force provided by the engine be F .

Resistive force $D \propto v^2 \Rightarrow D = kv^2$

At constant speed, $F = D$

Power output $P = Fv$

Power output $P = Dv$
 $= kv^3$

$$\Rightarrow 23 \text{ kW} = k(20)^3 \text{ -----(1)}$$

$$P = k(40)^3 \text{ -----(2)}$$

$$\frac{(2)}{(1)} \quad \frac{P}{23} = \frac{k(40)^3}{k(20)^3}$$

$$P = 184 \text{ kW}$$

8. Answer: B